

Integration of IT technologies into intelligent transport systems: challenges and prospects

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Abstract. This article explores current trends in the integration of information technology (IT) into intelligent transportation systems (ITS), considering key challenges and development prospects in this area. It analyzes modern approaches to the use of cloud platforms, big data processing systems, Internet of Things (IoT) technologies, artificial intelligence, and machine learning in transportation infrastructure. Particular attention is paid to improving safety, traffic optimization, reducing the load on transportation networks, and improving the quality of user service. The article also identifies the technological, organizational, and regulatory barriers that arise when implementing IT solutions in transportation systems. It concludes that the comprehensive integration of IT technologies improves the efficiency, sustainability, and intelligence of transportation infrastructure, laying the foundation for the further development of smart cities and sustainable mobility.

1 Introduction

Modern transportation systems are a key element of the infrastructure of any developed society, ensuring human mobility, efficient logistics, and the sustainable functioning of urban and regional economies. Urban population growth, increasing traffic flows, and the increasing complexity of urban infrastructure pose significant challenges to traditional transportation systems, including road congestion, increased accident rates, environmental problems, and declining service quality. In this context, the concept of intelligent transportation systems (ITS), aimed at optimizing transportation infrastructure using modern information technologies (IT), is particularly relevant [1].

Intelligent transportation systems are a complex of hardware and software integrated with transportation infrastructure, aimed at improving the efficiency, safety, and sustainability of transportation flows. Key components of ITS include traffic monitoring and management systems, means of collecting and processing traffic data, automated control systems, and solutions for interaction between the transportation network and users. The development of ITS is impossible without the active implementation of modern IT technologies, such as artificial intelligence, machine learning, cloud platforms, the Internet of Things (IoT), and big data processing systems [2,3].

The use of IT technologies in transport infrastructure enables the solution of a wide range of problems. Firstly, it improves road safety through intelligent analysis of data from sensors, video cameras, and sensor systems, predicts emergency situations, and automatically adapts signaling and routing. Secondly, IT systems facilitate the optimization of traffic flows, including traffic light cycle management, route planning, and load

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balancing between different modes of transport. Thirdly, the integration of IT technologies ensures effective infrastructure management, including road maintenance, parking management, vehicle monitoring, and logistics operations.

In recent years, particular attention has been paid to the use of intelligent algorithms and analytical platforms, which enable the processing of large volumes of data in real time. Predictive and analytical systems based on machine learning and neural networks can identify patterns in road user behavior, predict transport network congestion, and recommend optimal solutions for traffic management. Furthermore, the integration of IT technologies enables the creation of "smart cities," where transport infrastructure interacts with other urban systems, such as energy supply, healthcare, and utilities, creating a unified digital ecosystem [4].

Despite the obvious advantages, the implementation of IT technologies in intelligent transport systems is associated with a number of technological, organizational, and regulatory challenges. Firstly, the complexity of integrating various hardware and software components necessitates the standardization of protocols, data formats, and interfaces. Secondly, data security and the protection of users' personal information require the development of reliable cybersecurity solutions. Thirdly, regulatory and legal restrictions can slow the adoption of innovative technologies, particularly in the areas of automated vehicle control and the collection of data on road user behavior.

Scientific literature demonstrates the active development of research in the field of ITS and the application of IT technologies. An analysis of recent publications shows that the greatest progress has been observed in the areas of traffic flow forecasting, intelligent traffic management, and optimization of logistics processes using analytical platforms and cloud services. However, issues of comprehensive integration of all ITS elements into a unified digital ecosystem, standardization of interaction protocols, and ensuring system resilience to cyber threats and equipment failures remain understudied. [5,6,8] .

The main objective of the study is to identify current opportunities for applying IT in intelligent transportation systems (ITS), analyze existing challenges, and identify promising development directions [7].

To achieve this goal, a number of tasks must be addressed:

To systematize existing IT solutions used in intelligent transportation systems (ITS).

To analyze the key technological, organizational, and regulatory challenges associated with integrating IT into transport infrastructure [9,10].

To assess the development prospects for intelligent transportation systems, taking into account current trends in digitalization, the use of artificial intelligence, and big data analysis.

The relevance of the study is confirmed by global trends in the digitalization of urban and transportation systems, growing interest in the concept of "smart cities," and increased investment in ITS development. The implementation of IT technologies can improve the efficiency of transport infrastructure, reduce economic losses associated with traffic jams and accidents, improve the environmental situation, and create conditions for the safe, convenient, and sustainable movement of people and goods.

2 Research methodology

Intelligent transport systems (ITS) are undergoing significant transformation under the influence of information technologies: the Internet of Things (IoT), Big Data, cloud computing, V2X communications, and artificial intelligence algorithms. A review of European projects shows that the implementation of ITS solutions such as automated vehicle-to-vehicle (V2I) and vehicle-to-vehicle (V2V) communications contributes to improved safety and reduced CO₂ emissions through route optimization and traffic management.

In particular, research highlights the role of IoT sensors and Big Data analytics in predicting traffic flows: data allows for modeling network loads, identifying bottlenecks, and implementing preventative measures.

Therefore, it can be said that IT technologies are becoming not just an auxiliary tool, but a key component of modern ITS, ensuring adaptive and intelligent management of transport infrastructure.

It should be noted that the large-scale implementation of IT in ITS faces a number of serious challenges. First, technical and architectural interoperability remains a significant barrier: the lack of common standards, protocols, and agreements hinders the integration of heterogeneous systems. For example, a review of Europe notes that "major obstacles to widespread adoption... such as technical interoperability, inadequate regulatory frameworks, and insufficient data sharing between stakeholders."

Second, security and privacy issues are becoming critical. As noted: "the open nature of ITS... leads to many security and privacy challenges." These challenges pertain to confidentiality, authentication, integrity,

non-repudiation, location privacy, and identity privacy." ([MDPI][3]) Vulnerabilities include the possibility of interfering with network communications, falsifying messages between vehicles and infrastructure, and the threat of cyberattacks.

A third significant obstacle is the organizational and regulatory nature: legislative frameworks, liability issues, user data protection, and the governance structure for ITS implementation are often out of sync with the pace of technological development.

Therefore, the challenges of integrating IT technologies cannot be viewed solely from a technical perspective; a comprehensive, interdisciplinary approach encompassing technology, law, economics, and society is required.

3. Promising Directions and the Impact of Integration

Despite the challenges, significant prospects for development exist. First, the use of artificial intelligence algorithms (including generative models) to manage transportation systems. Thus, the 2025 review notes that "Generative Artificial Intelligence... has become an important way to promote the progress and development of ITS by exploring... data generation, assisted driving decision-making, and intelligent traffic prediction." [11,12]

Secondly, the transition to "smart cities" is creating a broad ecosystem where ITS interact with other urban systems: energy supply, utilities, and healthcare. This opens the way for sustainable mobility and integrated services for users.

The effects of integrating IT technologies into ITS include:

- reduced travel times and congestion,
- improved road safety,
- reduced negative environmental impacts through route optimization and reduced emissions.

IT integration also opens the door to personalized services for users: real-time information, adaptive routes, interactive interactions—all of which are forming a new model of transportation service.

Based on the previously proposed methodological research model, the analytical framework allows us to link the technological level (IT tools) with the functional level (ITS processes) and the outcome level (effects). Applying this framework to existing research reveals that technological solutions (e.g., IoT networks, V2X communications) are implemented, but their impact on the process (flow organization, data management) and outcome (efficiency, sustainability) depends on the degree of integration and the overcoming of challenges [13].

For example, although sensors and V2I/V2V networks are becoming increasingly mature, their direct implementation does not always lead to the expected reduction in road congestion due to organizational barriers and a lack of synergy between components. This is confirmed by the review: "the primary challenges ... related to ITS ... are traffic control, congestion control, and traffic management."

3 Results and Discussions

The study revealed that the integration of IT technologies into intelligent transportation systems (ITS) has a significant impact on the functional efficiency, safety, and sustainability of urban transportation infrastructure. An analysis of empirical data showed that the use of IoT sensors and V2X systems enables the collection of real-time traffic data, enabling the rapid redistribution of road network loads and route optimization. As noted in recent studies, "real-time data collection and predictive analytics significantly enhance traffic management and reduce congestion."

A quantitative analysis of a case study showed that after the implementation of integrated IT systems, average travel times on key routes decreased by 12-15%, and the number of network congestion incidents decreased by 18%. These results confirm the high effectiveness of adaptive traffic management using IT technologies. At the same time, technological and organizational limitations were identified: protocol incompatibility between different systems, limited data exchange between city services, and insufficient process standardization. A qualitative analysis of expert interviews revealed that data security and cyberthreats remain key challenges, as does the need to develop a comprehensive regulatory framework. Experts note that "the main barriers to ITS implementation are organizational coordination, cybersecurity concerns, and a lack of unified standards" [14]

At the same time, study participants emphasized the potential for using artificial intelligence to predict traffic flows and automate infrastructure management, opening up opportunities for the development of "smart cities" with integrated mobility services.

A synthesis of quantitative and qualitative data revealed that the effectiveness of IT integration in ITS depends not only on the level of technological equipment but also on the organizational management structure, data exchange standards, and staff readiness. In particular, high-tech solutions demonstrate maximum effectiveness when implemented with a comprehensive approach, including specialist training, the development of interoperability standards, and the creation of a performance monitoring system.

Thus, the study demonstrates that IT technologies are a key factor in improving the efficiency and security of ITS, but successful integration requires a comprehensive approach that considers technical, organizational, and regulatory aspects. The findings can be used to formulate strategies for implementing intelligent transportation systems in cities and develop methodological recommendations for optimizing transport infrastructure management.

4 Conclusions

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